Real-Life Data Investigation: A Healthy School Week

The **Eco and Wellness Club** at a school is investigating students' healthy habits. They collected data from students during one school week. Your task is to represent and analyse the data using **four different types of graphs**.

Data collected

A. Favourite Healthy Snack

Healthy Snack	Number of Students
Fruit	30
Sandwich	20
Nuts	15
Yoghurt	25
Salad	10

B. Books Read During the Week

Books Read	Number of Students
0-2	5
3-5	10
6-8	15
9-11	7
12-14	3

C. Water Intake

The number of glasses of water consumed by 20 students in one day was:

4, 5, 6, 5, 7, 6, 5, 8, 6, 4, 7, 5, 6, 7, 5, 6, 8, 5, 7, 6

D. Books Read by the Whole Class

The class recorded the total number of books read each month:

Month	**Books Read**
January	20
February	30
march	40
April	25
May	35

Task: Using the information above:

1. Bar Graph

Represent the **favourite healthy snacks** using a **bar graph**.

- Draw and label the X-axis and Y-axis.
- Choose a suitable scale.
- Use equal intervals.
- Give your graph a suitable title.

2. Histogram

Represent the **number of books read by students** using a **histogram**.

- X-axis → **Number of Books**
- Y-axis → **Frequency**
- Use equal intervals on the Y-axis.
- Remember: **Histogram bars must touch.**
- Give your histogram a suitable title.

3. Line Plot

Represent the **water intake of the 20 students** using a **line plot (dot plot)**.

- Draw a number line from **4 to 8**.
- Label the X-axis.
- Put an **X** for each student's result.
- Stack the Xs when values repeat.

- Give your line plot a suitable title.

4. Pictogram

Represent the **number of books read by the whole class each month** using a **pictogram**.

Key: 1 book = 5 books

- Choose the correct number of symbols for each month.
- Include a clear key.
- Give your pictogram a title.

Analyse Your Data

After completing all four graphs, answer:

1. Which healthy snack is the most popular?
2. How many students chose either fruit or yoghurt?
3. Which book-reading interval has the highest frequency?
4. How many students read **6 or more books**?
5. How many students drank **6 glasses of water**?
6. In which month did the class read the most books?
7. What is the difference between the highest and lowest monthly book totals?
8. **Which graph was the most suitable for showing each type of data? Explain why.**